

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

Objective1

Program:

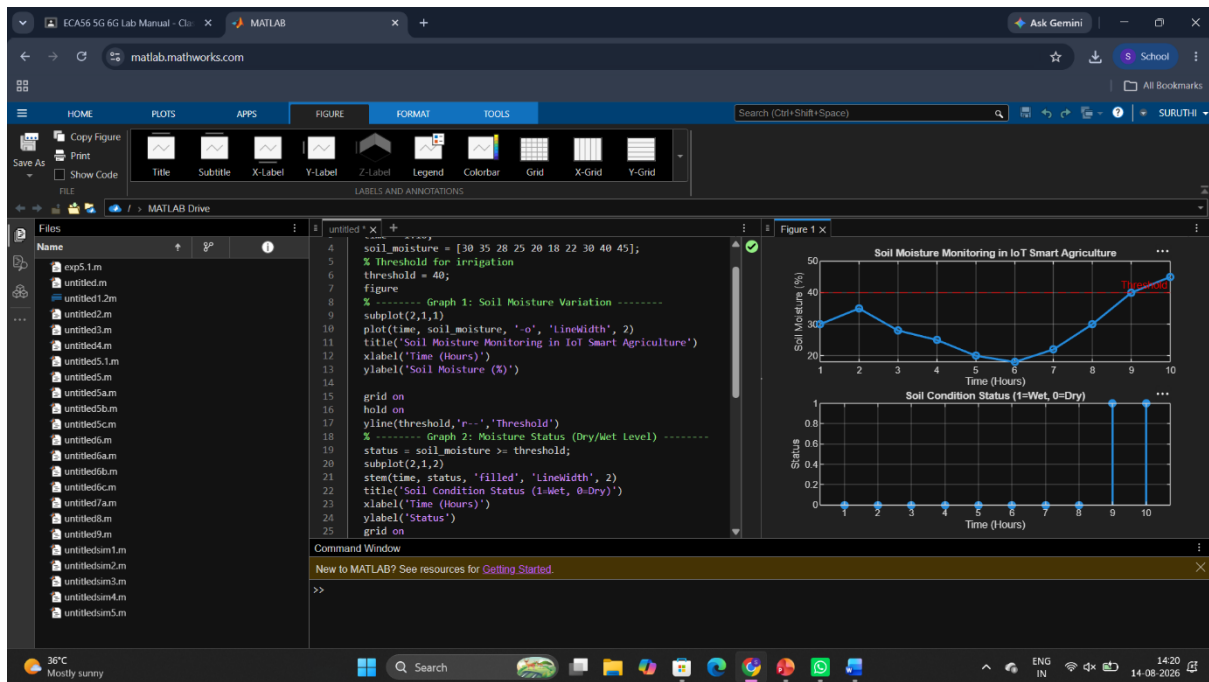
```
clc; clear; close all;

% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;

figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
...yline(threshold,'r--','Threshold')

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on

output:
```



Objective2:

Program:

clc; clear; close all;

% Simulated soil moisture (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)

pump_status = soil_moisture < threshold;

figure

% ----- Graph 1: Soil Moisture vs Pump Status -----

...hold on

ylines(threshold, 'r--', 'Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----

subplot(2,1,2)

```
stem(time, pump_status, 'filled','LineWidth',2)
```

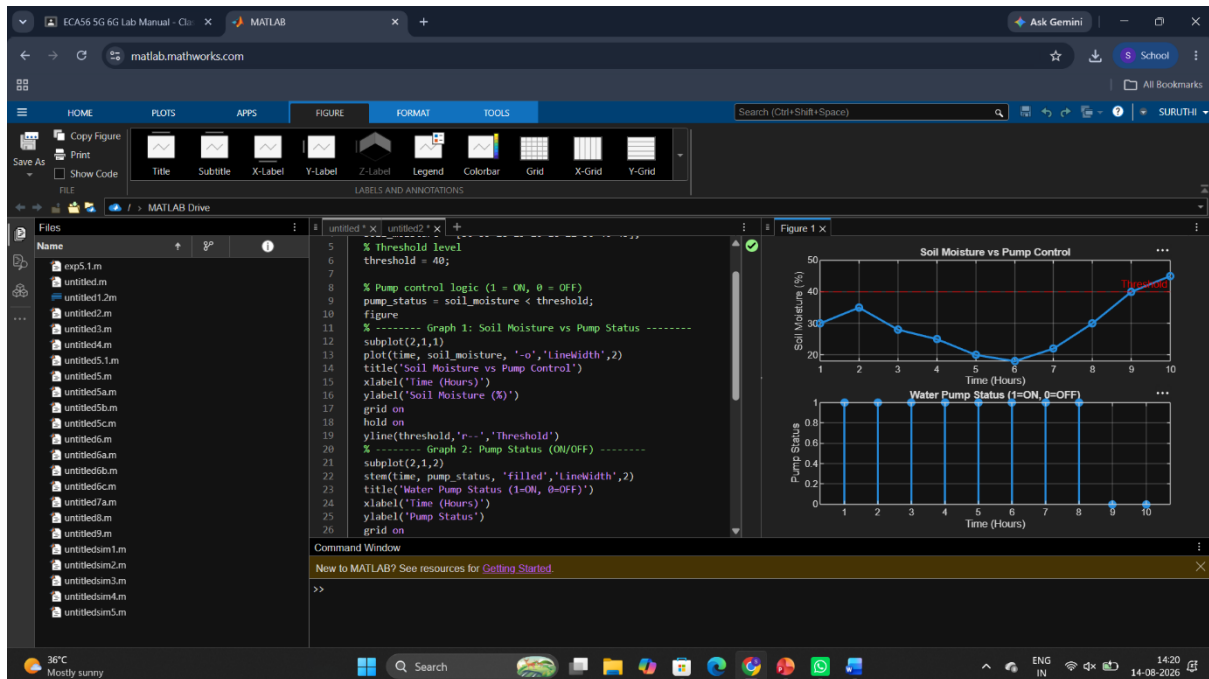
```
title('Water Pump Status (1=ON, 0=OFF)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Pump Status')
```

```
grid on
```

output:



Objective3:

Program:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Irrigation efficiency calculation (%)
```

```
efficiency = (soil_moisture ./ threshold) * 100;
```

```
efficiency(efficiency > 100) = 100; % limit to 100%
```

figure

...grid on

hold on

yligne(threshold,'r--','Threshold')

% ----- Graph 2: Irrigation Efficiency -----

subplot(2,1,2)

plot(time, efficiency, '-s','LineWidth',2)

title('Irrigation Efficiency (%) in Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Efficiency (%)')

grid on

output:

